

CONSTANTS OF FUNCTIONAL EQUATIONS OF L-FUNCTIONS

2.2–3.8

For $s \in \mathbb{C}$, write ω_s for the quasi-character

$$x \mapsto \|x\|^s$$

from K^* to $\mathbb{C}^*$.

If K is nonarchimedean, i.e. $K \neq \mathbb{R}, \mathbb{C}$, write $\mathcal{O}$ for the valuation ring of the valuation v of K , π for a uniformizer, k for the residue field $\mathcal{O}/(\pi)$, p for the characteristic of k , and put

$$d = [k : \mathbb{F}_p], \quad q = p^d = \#k.$$

Thus

$$\|x\| = q^{-v(x)}.$$

2.2.2. Suppose K is nonarchimedean. Let $\overline{K}$ be a separable closure of K , write $\overline{\mathcal{O}}$ for the integral closure of $\mathcal{O}$ in $\overline{K}$, let $\overline{k}$ be the residue field of $\overline{\mathcal{O}}$, an algebraic closure of k , and let v be the valuation of $\overline{K}$, with values in $\mathbb{Q}$, extending that of K .

There is a three-step filtration of the Galois group

$$\mathrm{Gal}(\overline{K}/K) \supset I \supset P$$

whose successive quotients are

$$\begin{aligned} \mathrm{Gal}(\overline{K}/K)/I &\xrightarrow{\sim} \mathrm{Gal}(\overline{k}/k) \xrightarrow{\sim} \widehat{\mathbb{Z}}, \\ I/P &\xrightarrow{\sim} \widehat{\mathbb{Z}}(1)(\overline{k}), \\ P &: \text{ a pro-}p\text{-group.} \end{aligned}$$

It is obtained as follows; for proofs we refer to [10].

- a) By transport of structure, $\mathrm{Gal}(\overline{K}/K)$ acts on $\overline{\mathcal{O}}$ and on $\overline{k}$. The inertia group I is the kernel of the reduction map

$$v' : \mathrm{Gal}(\overline{K}/K) \longrightarrow \mathrm{Gal}(\overline{k}/k).$$

The quotient $\mathrm{Gal}(\overline{k}/k)$ is canonically isomorphic to $\widehat{\mathbb{Z}}$, with topological generator the Frobenius substitution

$$\varphi : x \mapsto x^q.$$

- b) The wild inertia group P is the unique pro- p -Sylow subgroup of I . The equivariant morphism

$$t : I \longrightarrow \widehat{\mathbb{Z}}(1)(\overline{k}),$$

with kernel P , is characterized by the congruence, modulo the maximal ideal of $\overline{\mathcal{O}}$,

$$\frac{\sigma(x)}{x} = t(\sigma) v(x) \quad (\sigma \in I, x \in \overline{K}).$$

For a prime $\ell \neq p$, write t_ℓ for the composite

$$I \longrightarrow \widehat{\mathbb{Z}}(1)(\overline{k}) \longrightarrow \mathbb{Z}_\ell(1)(\overline{k}).$$

2.2.3. Let k be a finite field with q elements, and let $\overline{k}$ be its algebraic closure. Let $W(\overline{k}/k)$ be the subgroup of $\mathrm{Gal}(\overline{k}/k) \simeq \widehat{\mathbb{Z}}$ generated by

$$\varphi : x \mapsto x^q.$$

There is a canonical isomorphism

$$W(\overline{k}/k) \simeq \mathbb{Z}$$

with generator φ . The geometric Frobenius $F \in W(\overline{k}/k)$ is, by definition, φ^{-1} .

2.2.4. Let $K, \bar{K}$ be as in 2.2.2. The Weil group $W(\bar{K}/K)$, or absolute Weil group of K , is the subgroup of $\text{Gal}(\bar{K}/K)$ consisting of the σ such that $v'(\sigma)$ is an integral power of Frobenius φ . It is endowed with the topology inducing the natural topology on I , and for which I is open. The group $\text{Gal}(\bar{K}/K)$ is the completion of $W(\bar{K}/K)$ for the topology of open subgroups of finite index; if L is a finite extension of K in $\bar{K}$, the corresponding subgroup of $W(\bar{K}/K)$ is identified with $W(\bar{K}/L)$. We shall sometimes call geometric Frobenius elements those elements of $\text{Gal}(\bar{K}/K)$ or $W(\bar{K}/K)$ whose image is F .

2.2.5. Suppose K is archimedean, with algebraic closure $\bar{K}$. Define $W(\bar{K}/K)$ as the following extension of $\text{Gal}(\bar{K}/K)$ by $\bar{K}^*$:

- 1) $K \simeq \mathbb{R}$: $W(\bar{K}/K)$ is generated by $\bar{K}^*$ and an element F , with

$$F^2 = -1, \quad FzF^{-1} = \bar{z} \quad (z \in \bar{K}^*).$$

- 2) $K \simeq \mathbb{C}$: $W(\bar{K}/K) = \bar{K}^*$.

2.3. Local class field theory provides an isomorphism between $W(\bar{K}/K)^{\text{ab}}$ and K^* .

Assume K nonarchimedean. For a reason explained in 3.6, we normalize this isomorphism, i.e. choose it or its inverse, so that geometric Frobenius elements correspond to uniformizers:

$$\begin{array}{ccccccc} P & \hookrightarrow & I & \hookrightarrow & W(\bar{K}/K) & \longrightarrow & W(\bar{k}/k) \simeq \mathbb{Z} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow -1 \\ 1 + (\pi) & \hookrightarrow & \mathcal{O}^* & \hookrightarrow & K^* & \xrightarrow{v} & \mathbb{Z}. \end{array}$$

A representation of $W(\bar{K}/K)$ is unramified, respectively tamely ramified, if it is trivial on I , respectively on P . Similarly, a quasi-character χ of K^* is unramified, respectively tamely ramified, if it is trivial on $\mathcal{O}^*$, respectively on $1 + (\pi)$. For ℓ prime to p , the action of $\text{Gal}(\bar{K}/K)$ on $\mathbb{Z}_\ell(1)$ is unramified, described by the quasi-character

$$\omega_1 = \|x\|$$

of K^* .

For K complex, the isomorphism of local class field theory is the evident one on 2.2.5; for K real, it is the one which makes 2.3.1 and 2.3.2 below true.

Let $L \subset \bar{K}$ be a finite extension of K , so that $W(\bar{K}/L) \subset W(\bar{K}/K)$. The following diagrams, where N is the norm and t the transfer, are commutative ([10], XIII, §4):

$$\begin{array}{ccccc} W(\bar{K}/L) & \longrightarrow & W(\bar{K}/L)^{\text{ab}} & \xrightarrow{\sim} & L^* \\ \downarrow & & \downarrow & & \downarrow N_{L/K} \\ W(\bar{K}/K) & \longrightarrow & W(\bar{K}/K)^{\text{ab}} & \xrightarrow{\sim} & K^*, \end{array} \tag{2.3.1}$$

$$\begin{array}{ccc} W(\bar{K}/K)^{\text{ab}} & \xrightarrow{\sim} & K^* \\ t \downarrow & & \downarrow \\ W(\bar{K}/L)^{\text{ab}} & \xrightarrow{\sim} & L^*. \end{array} \tag{2.3.2}$$

2.4. Let K be a function field in one variable over $\mathbb{F}_p$, and let k be its field of constants, the algebraic closure of $\mathbb{F}_p$ in K . Let $\bar{K}$ be a separable closure of K , let $\bar{k}$ be the algebraic closure of k in $\bar{K}$, and let $\text{Gal}^0(\bar{K}/K)$ be the kernel of the restriction map from $\text{Gal}(\bar{K}/K)$ to $\text{Gal}(\bar{k}/k)$. The absolute global Weil group $W(\bar{K}/K)$ is, by analogy with 2.2.4, defined by the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Gal}^0(\bar{K}/K) & \longrightarrow & W(\bar{K}/K) & \longrightarrow & W(\bar{k}/k) \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \text{Gal}^0(\bar{K}/K) & \longrightarrow & \text{Gal}(\bar{K}/K) & \longrightarrow & \text{Gal}(\bar{k}/k) \longrightarrow 0. \end{array}$$

Let v be a place of K , and suppose a place of $\overline{K}$ above v has been chosen. There is then a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & I_v & \longrightarrow & W(\overline{K}_v/K_v) & \longrightarrow & W(\overline{k}_v/k_v) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \text{Gal}^0(\overline{K}/K) & \longrightarrow & W(\overline{K}/K) & \longrightarrow & W(\overline{k}/k) \longrightarrow 0. \end{array}$$

Global class field theory provides an isomorphism

$$W(\overline{K}/K)^{\text{ab}} \xrightarrow{\sim} \mathbb{A}_K^*/K^*$$

where $\mathbb{A}_K^*/K^*$ is the idèle-class group, making the diagrams

$$\begin{array}{ccc} W(\overline{K}_v/K_v)^{\text{ab}} & \longrightarrow & W(\overline{K}/K)^{\text{ab}} \\ \downarrow & & \downarrow \\ K_v^* & \longrightarrow & \mathbb{A}_K^*/K^* \end{array}$$

commutative.

2.5. For the definition of global Weil groups in the case of number fields, we refer to Weil [19] or to [2], XIV. We shall use them only incidentally.

3. Reminders on L -functions

3.1. Let K be a local field and use again the notation of 2.2.1. We denote by dx a Haar measure on K , by d^*x a Haar measure on K^* , and by ψ a nontrivial additive character of K .

3.2. For a quasi-character, i.e. a continuous homomorphism,

$$\chi : K^* \longrightarrow \mathbb{C}^*,$$

define $L(\chi) \in \mathbb{C} \cup \{\infty\}$ as follows.

$$K \simeq \mathbb{R}. \tag{3.2.1}$$

Put

$$\Gamma_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(s/2),$$

and, for x the embedding of K in $\mathbb{C}$ and $N = 0$ or 1 ,

$$L(x^{-N} \omega_s) = \Gamma_{\mathbb{R}}(s).$$

$$K \simeq \mathbb{C}. \tag{3.2.2}$$

Put

$$\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s} \Gamma(s),$$

and, for z an embedding of K in $\mathbb{C}$ and $N \geq 0$,

$$L(z^{-N} \omega_s) = \Gamma_{\mathbb{C}}(s).$$

$$K \text{ nonarchimedean.} \tag{3.2.3}$$

Put

$$L(\chi) = 1 \quad (\chi \text{ ramified}), \quad L(\chi) = \frac{1}{1 - \chi(\pi)} \quad (\chi \text{ unramified}).$$

3.3. Given ψ and dx , put, for f a C^∞ function of compact support on K ,

$$\widehat{f}(y) = \int_K f(x) \psi(xy) dx.$$

Define $\varepsilon(\chi, \psi, dx) \in \mathbb{C}^*$ by Tate's local functional equation, independent of f ,

$$\frac{\int_{K^*} \widehat{f}(x) \omega_1 \chi^{-1}(x) d^*x}{L(\omega_1 \chi^{-1})} = \varepsilon(\chi, \psi, dx) \frac{\int_{K^*} f(x) \chi(x) d^*x}{L(\chi)}. \quad (3.3.1)$$

The same Haar measure d^*x is used on both sides. Both sides are defined by analytic continuation in χ in the families $\chi \omega_s$, $s \in \mathbb{C}$. In these families, ε is of the form Ae^{Bs} .

One has

$$\varepsilon(\chi, \psi, a dx) = a \varepsilon(\chi, \psi, dx), \quad (3.3.2)$$

$$\varepsilon(\chi, \psi(ax), dx) = \chi(a) \|a\|^{-1} \varepsilon(\chi, \psi, dx). \quad (3.3.3)$$

3.4. For K nonarchimedean, put:

$$n(\psi) = \text{the largest integer } n \text{ such that } \psi|_{\pi^{-n}\mathcal{O}} = 1;$$

$$a(\chi) = \begin{cases} 0, & \chi \text{ unramified,} \\ \text{the least } m \text{ such that } \chi|_{1+(\pi)^m} = 1, & \chi \text{ ramified;} \end{cases}$$

$$sw(\chi) = \begin{cases} 0, & \chi \text{ unramified or tamely ramified,} \\ a(\chi) - 1, & \chi \text{ ramified;} \end{cases}$$

and let γ be an element of K^* of valuation

$$v(\gamma) = n(\psi) + sw(\chi) + 1.$$

The local constant ε is given by (3.3.2), (3.3.3), and the following formulas.

$$K \simeq \mathbb{R}. \quad (3.4.1)$$

Let x be the embedding of K in $\mathbb{C}$, and $N = 0$ or 1 . For $\psi = \exp(2\pi i x)$ and dx the Lebesgue measure,

$$\varepsilon(x^{-N} \omega_s, \psi, dx) = i^N.$$

$$K \simeq \mathbb{C}. \quad (3.4.2)$$

Let z be an embedding of K in $\mathbb{C}$, and $N \geq 0$. For

$$\psi = \exp(2\pi i \operatorname{Tr}_{\mathbb{C}/\mathbb{R}}(z))$$

and

$$dx = |dz \wedge d\bar{z}| = 2 da db \quad (z = a + bi),$$

one has

$$\varepsilon(z^{-N} \omega_s, \psi, dx) = i^N.$$

3.4.3. K nonarchimedean. If χ is unramified,

$$\int_{\mathcal{O}} dx = 1, \quad n(\psi) = 0,$$

then

$$\varepsilon(\chi, \psi, dx) = 1. \quad (3.4.3.1)$$

For χ ramified, one has

$$\varepsilon(\chi, \psi, dx) = \int_{K^*} \chi^{-1}(x) \psi(x) dx = \sum_n \int_{v(x)=n} \chi^{-1}(x) \psi(x) dx = \int_{\gamma^{-1}\mathcal{O}^*} \chi^{-1}(x) \psi(x) dx. \quad (3.4.3.2)$$

We shall see that it is natural to unify these two formulas as follows. If $\varepsilon_0(\chi, \psi, dx)$ is defined by

$$\varepsilon(\chi, \psi, dx) = \varepsilon_0(\chi, \psi, dx) \quad (\chi \text{ ramified}),$$

and

$$\varepsilon(\chi, \psi, dx) = \varepsilon_0(\chi, \psi, dx) \chi(\pi)^{-1} \quad (\chi \text{ unramified}), \quad (3.4.3.3)$$

then

$$\varepsilon_0(\chi, \psi, dx) = \int_{\gamma^{-1}\mathcal{O}^*} \chi^{-1}(x) \psi(x) dx. \quad (3.4.3.4)$$

From (3.4.3.3) and (3.4.3.4), one deduces that, for ω unramified,

$$\varepsilon_0(\chi\omega, \psi, dx) = \omega(\pi)^{n(\psi)+sw(\chi)+1} \varepsilon_0(\chi, \psi, dx), \quad (3.4.3.5)$$

and

$$\varepsilon(\chi\omega, \psi, dx) = \omega(\pi)^{n(\psi)+a(\chi)} \varepsilon(\chi, \psi, dx).$$

3.5. Suppose K nonarchimedean, and resume the notation of 2.2.1. Let V be a finite-dimensional vector space over a field E of characteristic 0, or over $\mathbb{C}$. The representations

$$\rho : W(\overline{K}/K) \longrightarrow \mathrm{GL}(V)$$

will always be assumed continuous: an open subgroup of I acts trivially. For V the space of a representation and V^I the subspace of inertia invariants, put

$$Z(V; t) = \det(1 - Ft^d, V^I)^{-1}, \quad (3.5.1)$$

$$L(V) = Z(V; 1) \quad (E \in E^* \cup \{\infty\}). \quad (3.5.2)$$

Let

$$\omega^t : W(\overline{K}/K) \longrightarrow E((t))^*$$

be the unramified character such that $\omega^t(F) = t^d$. In an evident sense, $\omega_s = \omega^{p^{-s}}$. One has

$$Z(V, t) = L(V \otimes \omega^t). \quad (3.5.3)$$

3.6. It is so that (3.2.3) and (3.5.2) are compatible that we normalized the local class field theory isomorphism so that geometric Frobenius elements correspond to uniformizers. Both (3.2.3) and (3.5.2) are very natural: 3.2.3 is forced from the analytic point of view (cf. 3.3.1), and 3.5.2 follows the tradition of geometers; it is geometric Frobenius which tends to act on the cohomology of algebraic varieties with algebraic integers as eigenvalues.

3.7. The irreducible continuous complex representations of $W(\overline{K}/K)$ are, for K complex, the quasi-characters. For K real, besides the quasi-characters of K^* , they are the representations induced by quasi-characters χ of $\overline{K}^*$ such that $\chi \neq \chi \circ \sigma$, where σ is complex conjugation of $\overline{K}$.

For any complex representation V , expressed in the Grothendieck group of representations of $W(\overline{K}/K)$ as a sum of simple representations, define $L(V)$ as follows:

$$K \simeq \mathbb{R}, \quad V = \sum_i [\chi_i] + \sum_j \text{Ind}_{\overline{K}^*}^W(\chi_j), \quad (3.7.1)$$

$$L(V) = \prod_i L_K(\chi_i) \prod_j L_{\overline{K}}(\chi_j).$$

$$K \simeq \mathbb{C}, \quad V = \sum_i [\chi_i], \quad L(V) = \prod_i L(\chi_i). \quad (3.7.2)$$

Proposition 3.8. Let K be a local field and $\overline{K}$ an algebraic closure of K .

(i) For every exact sequence

$$0 \longrightarrow V' \longrightarrow V \longrightarrow V'' \longrightarrow 0$$

of complex representations of $W(\overline{K}/K)$, one has

$$L(V) = L(V')L(V''). \quad (3.8.1)$$

(ii) Let L be a finite separable extension of K in $\overline{K}$, and let V_L be a complex representation of $W(\overline{K}/L)$. Let V_K be the representation induced to $W(\overline{K}/K)$. Then

$$L(V_K) = L(V_L). \quad (3.8.2)$$

Assertion (i) is trivial. In the archimedean case, assertion (ii) follows from the duplication formula

$$\Gamma_{\mathbb{C}}(s) = \Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1),$$

corresponding to

$$\text{Ind}([\omega_s]) = [\omega_s] + [\chi^{-1}\omega_{s+1}].$$

We prove (ii) for K nonarchimedean. Let k and ℓ be the residue fields of K and L . Then

$$V_K^{I_K} = \left(\text{Ind}_{W(\overline{K}/L)}^{W(\overline{K}/K)} V_L \right)^{I_K} \simeq \text{Ind}_{W(\overline{\ell}/\ell)}^{W(\overline{k}/k)} (V_L^{I_L})$$

as $W(\overline{k}/k)$ -representations, both sides being identified with

$$\text{Hom}_{W(\overline{K}/L)}(W(\overline{k}/k), V_L),$$

the covariant functions from $W(\overline{k}/k)$ to V_L , $W(\overline{K}/L)$ acting on $W(\overline{k}/k)$ by translation; cf. the commutative diagram